# A micro Lie theory: the review deck

Solà, Deray and Atchuthan, arXiv:1812.01537v9

Lesson t2.6 · equation to code, with the traps

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# What the deck covers

1. Exp and Log on SO(3), $S^3$, SE(2)
2. Jacobian blocks and the chain rule
3. ESKF and Gauss-Newton SAM
4. Measured numbers
5. Mistakes already made
6. Implementation plan

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# SO(3) Exp and Log (App. B, pp. 14–15)

$$\mathbf{R} = \mathbf{I} + \sin\theta\,[\mathbf{u}]_\times + (1-\cos\theta)[\mathbf{u}]_\times^2 \tag{134}$$

$$\theta\mathbf{u} = \frac{\theta(\mathbf{R}-\mathbf{R}^\top)^\vee}{2\sin\theta} \tag{135}$$

**Trap: (135) is 0/0 at θ → 0 and at θ → π**

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# Robust coefficients and quaternion Log (C1)

```python
def coef_a(th):   # (1 - cos th)/th^2, half-angle form
    s = np.sinc(th / (2 * np.pi))     # sin(th/2)/(th/2)
    return 0.5 * s * s

def coef_b(th):   # (th - sin th)/th^3, series below 0.3
    th = abs(th)
    if th < 0.3:
        return sum((-1) ** k * th ** (2 * k) / math.factorial(2 * k + 3)
                   for k in range(5))
    return (th - np.sin(th)) / th ** 3

def quat_log(q):  # (133), w > 0 rule
    w, v = (q[0], q[1:]) if q[0] >= 0 else (-q[0], -q[1:])
    n = np.linalg.norm(v)
    return np.zeros(3) if n == 0.0 else 2 * np.arctan2(n, w) / n * v
```

- Half-angle form: no subtraction, no branch at θ = 0
- Flip q to w ≥ 0 before Log: the short rotation

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# Why the robust Log: Log(Exp(θ)) error

**NaN**

naive Log at $\theta = 10^{-9}$

**0.16 rad**

naive Log at $\theta = \pi - 10^{-7}$

**$6.4 \times 10^{-16}$**

robust Log at both ends, rad

- Coefficient errors for θ from $10^{-12}$ to π: $3.3 \times 10^{-16}$ for a and $4.0 \times 10^{-15}$ for b

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# Elementary blocks (III-B, p. 10)

$$\mathbf{J}_{\mathcal{X}}^{\mathcal{X}^{-1}} = -\mathbf{Ad}_{\mathcal{X}} \quad (62) \qquad \mathbf{J}_{\mathcal{X}}^{\mathcal{X} \circ \mathcal{Y}} = \mathbf{Ad}_{\mathcal{Y}}^{-1} \quad (65) \qquad \mathbf{J}_{\mathcal{Y}}^{\mathcal{X} \circ \mathcal{Y}} = \mathbf{I} \quad (66)$$

$$\mathbf{Ad}_{\mathrm{Exp}(\boldsymbol{\tau})} = \mathbf{J}_l(\boldsymbol{\tau})\mathbf{J}_r^{-1}(\boldsymbol{\tau}) \quad (75) \qquad \mathbf{J}_r(-\boldsymbol{\tau}) = \mathbf{J}_l(\boldsymbol{\tau}) \quad (76)$$

**Trap: mixing right and left $\ominus$ leaves the Jacobian off by an Ad**

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# Log, $\oplus$ and $\ominus$ blocks (III-C, p. 11)

| Block | Result | Eq. |
|---|---|---|
| J of Log $\mathcal{X}$ w.r.t. $\mathcal{X}$ | $J_r^{-1}(\tau)$ | (79) |
| J of $\mathcal{X} \oplus \tau$ w.r.t. $\mathcal{X}$ | $Ad_{Exp(\tau)^{-1}}$ | (80) |
| J of $\mathcal{X} \oplus \tau$ w.r.t. $\tau$ | $J_r(\tau)$ | (81) |
| J of $\mathcal{Y} \ominus \mathcal{X}$ w.r.t. $\mathcal{X}$ | $-J_l^{-1}(\tau)$ | (82) |
| J of $\mathcal{Y} \ominus \mathcal{X}$ w.r.t. $\mathcal{Y}$ | $J_r^{-1}(\tau)$ | (83) |

**Near $\tau = 0$ all of them tend to $\pm I$; assuming that slows Gauss-Newton**

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# SE(2) Exp and Log with V(θ) (C4)

```python
def se2_V(th):                          # (158)
    return np.sinc(th/np.pi)*np.eye(2) + th*coef_a(th)*J1

def se2_exp(tau):                       # (156)
    M = np.eye(3); M[:2, :2] = so2(tau[2])
    M[:2, 2] = se2_V(tau[2]) @ tau[:2]
    return M

def se2_log(M):                         # (157)
    th = np.arctan2(M[1, 0], M[0, 0])
    A, B = np.sinc(th/np.pi), th*coef_a(th)
    rho = (A*np.eye(2) - B*J1) @ M[:2, 2] / (A*A + B*B)
    return np.r_[rho, th]
```

- $V^{-1} = (A \cdot I - B \cdot [1]_x)/(A^2 + B^2)$: no 2×2 inverse call
- $\tau = (\rho, \theta)$, translation first

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# SE(2) J_r third column: printed vs rewritten

## Printed (163)

- Terms with $\theta^2$ in the denominator
- Off by $2.7 \times 10^{-3}$ at $\theta = 2 \times 10^{-7}$
- Off by $1.4 \times 10^{-4}$ at $\theta = 10^{-6}$
- Lands in range when driving straight

## Rewritten with a(θ), b(θ)

- Same coefficients as SO(3): a, b
- Series below the switch
- Full precision at every θ
- manif's switch: $\theta = 1.5 \times 10^{-7}$

**On a robot, either live with manif's error or patch those two lines**

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# ESKF correction, robust form (C8)

```python
def eskf_update(X, P, y, b, Q_meas):          # (101)-(102)
    R, t = X[:2, :2], X[:2, 2]
    z = y - R.T @ (b - t)
    H = -np.hstack([np.eye(2), (R.T @ J1 @ (b - t))[:, None]])
    Z = H @ P @ H.T + Q_meas
    K = cho_solve(cho_factor(Z), H @ P).T     # no Z^-1
    dx = K @ z
    IKH = np.eye(3) - K @ H
    P = IKH @ P @ IKH.T + K @ Q_meas @ K.T    # Joseph form
    Jr = se2_Jr(dx)                           # reset, not in (102)
    P = Jr @ P @ Jr.T
    return X @ se2_exp(dx), 0.5 * (P + P.T)
```

- Gate on $z^T Z^{-1} z$ above 9.21 ($\chi^2$, 2 dof, 99 %)

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# ESKF consistency on manif's example

**3.03**

ANEES, Gaussian noise (3 is consistent)

**1.00**

ANEES, uniform noise

**6.8**

ANEES at the example's σ, with reset

**8.3**

same, without the reset

- manif's example draws uniform noise in $[-\sigma, \sigma]$ but tells the filter $\sigma^2$: use Gaussian in every consistency test

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# Gauss-Newton SAM on Fig. 12

19×1

**5**

stacked Jacobian J

**4**

iterations to converge

**481.4**

initial cost

**1.07**

final cost

- Solve (109) by Cholesky or QR on whitened residuals, never $(J^T J)^{-1}$; J is 33 % filled, $J^T J$ 60 %

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# Symptom to fix

| Symptom | Cause | Fix |
|---|---|---|
| FD slope 1, not 2 | Wrong block or $\ominus$ side | Test each block alone |
| Right at X = I only | Missing Ad | (46), (57); generic Ad (31) |
| NaN at rest | Divide by $\theta$ | Taylor branch; test $\theta = 10^{-10}$ |
| Error jumps by $2\pi$ | Branch cut, $w < 0$ | Flip q; wrap angles |
| Slow Gauss-Newton | $J_r^{-1} \approx I$ assumed | Use (82)–(83) in full |
| Overconfident after update | P not reset | $J_r(\delta x)$ reset, ref. [8] |

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# Mistakes already made

| Where | Error | Fix |
|---|---|---|
| Lecture L13 | Composite said not to be a Lie group | It is a direct product |
| Lecture L15 | Gain lines labelled (101) | Unnumbered, above (101) |
| T1 Q5 | $J_r^{-1}$ grows like $\theta^2$ past $\pi$ | $\theta/(2\sin(\theta/2))$; $\pi/2$ at $\pi$ |
| Fig. 12 citation | Given on p. 13 | It is p. 12 |
| Session 2.6d tile | floor($\log_{10}$) hid errors up to 10× | One-digit mantissa |
| Card C9 | Stale iteration count | 4 iterations, 481.4 → 1.07 |

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# Implementation plan: library first, then estimators

**1 SO(3), ops**

Exp/Log, J_r, J_l, ⊕, ⊖, Ad; check vs expm and just fd

**2 SE(2), SE(3)**

Rewritten a, b and series Q; check vs mpmath at small θ

**3 Compo site**

Diamond ⊕/⊖ over a (group, dim, offset) table

**4 ESKF, SAM**

(99)–(102), then (103)–(111); F, G, H and J by FD

**5 Real data**

UTIAS planar dataset, then the tracked robot; manif on the robot

**Correctness first: each step has an oracle that must pass before the next**

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